

Kishan — “RescueRank”

UAV-based room priority ranking for firefighter rescue support

Scope

Moderate and realistic if scoped as a software MVP. The project focuses on using images from different rooms, detecting humans, fire, and smoke, then ranking rooms based on rescue urgency. The system does not make the final rescue decision; it organizes visual evidence so firefighters can quickly review which rooms may need attention first.

Core pipeline:

1. Take images from different rooms.
2. Run object detection for:
 - humans
 - fire
 - smoke
3. Assign each room a rescue-priority score.
4. Rank rooms from highest to lowest urgency.
5. Display the ranked list and annotated images in a firefighter-facing dashboard.

This is feasible as an MVP in 12–13 days if the team uses public dataset images as simulated UAV room captures instead of trying to deploy a real UAV. The ranking logic can be rule-based at first, which keeps the system explainable and easy to demonstrate.

The main technical challenge is not the dashboard or ranking algorithm. The biggest challenge is getting reliable detection of humans, fire, and smoke from public datasets that may not perfectly match real fire-rescue conditions.

Subscriptions Required

Claude Max for building the system.

Recommended free/open-source stack:

- Python
- YOLO / Ultralytics for object detection
- OpenCV for image processing
- Streamlit for dashboard

- Pandas / NumPy for ranking and results table
- Roboflow, optional, for dataset formatting or annotation conversion
- Google Colab or Kaggle, optional, for model training if local GPU is unavailable

Possible cost:

- **No cost** if using pretrained models and public datasets.
- **Optional GPU cost** if training a custom fire/smoke model from scratch.
- **No UAV required for MVP** because the project can simulate UAV input using dataset images.

Bottom line: the software side is manageable. The team should avoid spending MVP time on real UAV navigation, live drone streaming, or hardware integration.

Dataset Availability

Dataset availability is the main risk, but it is manageable.

There is likely no single perfect dataset that contains all of these together:

- indoor room images
- UAV viewpoint
- fire
- smoke
- humans/victims
- room-level rescue priority labels

So the realistic approach is to combine datasets and manually create the room-priority structure.

Useful dataset types:

- Fire/smoke detection datasets for detecting fire and smoke.
- Indoor fire datasets for room-like scenes.
- General person-detection datasets or pretrained YOLO models for human detection.
- Manually grouped image folders to represent different rooms.

Best MVP data strategy:

Use fire/smoke datasets for fire and smoke detection, and use a pretrained YOLO person detector for humans. Then manually group images into “Room 1,” “Room 2,” “Room 3,” etc. Each group becomes one simulated UAV room observation.

Example structure:

dataset/
room_1/
image_1.jpg
image_2.jpg
room_2/
image_1.jpg
image_2.jpg
room_3/
image_1.jpg

The system should rank rooms, not just individual images.

Priority Ranking Logic

For the MVP, use a simple explainable score.

Priority Score =

$0.55 \times \text{Human Detection}$

$+ 0.25 \times \text{Fire Detection}$

$+ 0.15 \times \text{Smoke Detection}$

$+ 0.05 \times \text{Number of Humans Factor}$

Example ranking rules:

**Rank
Level**

Condition

Meaning

Highest	Human + fire/smoke detected	Immediate rescue concern
High	Human detected, no fire visible	Possible trapped person
Medium	Fire/smoke detected, no human	Hazardous room
Low	No human, no fire/smoke	Lowest immediate priority

Example output:

Rank 1: Room 3

Reason: Human detected + fire detected + smoke detected

Priority Score: 0.91

Rank 2: Room 1

Reason: Human detected, no visible fire

Priority Score: 0.63

Rank 3: Room 2

Reason: Fire detected, no human detected

Priority Score: 0.41

This makes the system easy to explain because every ranking has a clear reason behind it.

MVP Feasibility

Very feasible if scoped correctly.

Must-have MVP:

- Upload or select room images
- Detect person, fire, and smoke
- Generate annotated images
- Calculate room priority score
- Rank rooms
- Show results in a Streamlit dashboard

Should-have if time allows:

- Confidence threshold adjustment
- Multiple images per room
- “Reason for rank” explanation
- Export ranked report as CSV or PDF
- Low-confidence warning when detections are uncertain

Avoid for MVP:

- Real UAV deployment
- Real-time drone video stream
- Thermal camera integration
- Complex autonomous navigation
- Advanced multi-modal fusion
- Full firefighter operational validation

Biggest Risks

The biggest risk is **dataset mismatch**.

Public datasets may contain fire and smoke images, but they may not look exactly like UAV images from real indoor fire-rescue scenes. The team should state clearly:

For the MVP, UAV input is simulated using public image datasets. The project focuses on the computer vision and room-priority decision-support pipeline, not real UAV navigation.

Second risk: **person detection in smoke**.

A pretrained person detector may miss humans in smoky, blurry, dark, or partially occluded images. This should be shown as a limitation.

Third risk: **ranking is not the same as real rescue priority.**

The system can rank rooms based on visible cues, but it cannot fully understand building layout, structural collapse risk, oxygen levels, blocked paths, heat intensity, or firefighter safety conditions.

Bottom Line

This is a strong and achievable MVP because the goal is clear:

dataset room images → detect human/fire/smoke → calculate priority score → rank rooms → show annotated photos to firefighters.

The system should be presented as a **decision-support tool**, not an autonomous rescue system. Its main value is helping firefighters quickly organize and review room-level visual information during an emergency.

Quick Takeaways

- **Best project framing:** firefighter decision-support system, not autonomous rescue decision-maker.
- **Easiest MVP path:** public datasets + pretrained YOLO + Streamlit dashboard.
- **Biggest data issue:** no perfect UAV indoor fire-victim dataset exists.
- **Most important feature:** explainable room ranking with annotated images.
- **Core output:** ranked list of rooms with photos, detections, confidence scores, and reason for ranking.
- **Do not claim:** real-world deployment readiness.
- **Do claim:** prototype system for prioritizing room-level visual evidence for firefighter review.